

Yes. For **DMCT — Data Mining Concepts & Techniques (CS24303)**, I'll use the same **complete-master-notes format**, but DMCT needs more emphasis on **numericals, algorithms, comparison tables, worked examples, and active recall**.

I checked your uploaded **CS24303 syllabus/study guide** and the **DMCT PYQs**. Your syllabus has **46 topics across M1–M5**: 14 + 6 + 9 + 7 + 10.

Key Insight

Important: The PYQ labels below refer to questions actually present in your uploaded IT335/IT426 papers. IT335 is the directly relevant **B.Tech 5th-semester CSE/IT** paper; IT426 papers are additional DMCT papers from another semester/branch and are useful for pattern coverage.

DMCT COMPLETE MASTER NOTES

Data Mining Concepts & Techniques — CS24303

The course in one picture

```
RAW DATA
↓
UNDERSTAND DATA
↓
PREPROCESS
↓
DATA WAREHOUSE
↓
FIND PATTERNS
↓
CLASSIFY / PREDICT
↓
CLUSTER
↓
DISCOVER KNOWLEDGE
```

Think:

Key Insight

Data → Clean → Store → Mine → Evaluate → Knowledge

MODULE I — INTRODUCTION TO DATA MINING

14/14 topics

Your uploaded syllabus specifies these 14 topics: Introduction, Relational Databases, Data Warehouses, Transactional Databases, Advanced Database Systems and Applications, Data Mining Functionalities, Classification of Data Mining Systems, Major Issues, Data, Data Objects & Attribute Types, Statistical Descriptions, Visualization, Similarity and Dissimilarity.

1. Introduction to Data Mining

What is Data Mining?

Data mining is the process of discovering useful, previously unknown, and potentially valuable patterns from large datasets.

```
Large Dataset
  ↓
Data Mining
  ↓
Patterns
  ↓
Knowledge
  ↓
Decision
```

Example

A supermarket has millions of transactions.

Data mining may discover:

💡 Key Insight

Customers buying **bread + butter** frequently also buy **jam**.

That pattern can be used for:

- recommendations
- product placement
- promotions

Data Mining vs Database Query

Database Query	Data Mining
Retrieves known information	Discovers hidden patterns
User asks specific question	Pattern may be unknown
Exact result	Statistical/pattern-based result

2. Relational Databases

Data is represented using **tables**.

Example:

Student_ID	Name	Dept	CGPA
101	A	CSE	8.5
102	B	IT	7.9

Important concepts

- relation → table
- tuple → row

- attribute → column
- primary key → uniquely identifies tuple
- foreign key → references another relation

Why mine relational databases?

Because organizations already store huge amounts of structured data in relational systems.

3. Data Warehouses

A **data warehouse** is a centralized repository designed mainly for analysis and decision support.

Typical properties:

Subject-oriented

Organized around subjects such as:

- sales
- customers
- products

Integrated

Data from different sources is standardized.

Time-variant

Historical data is maintained.

Non-volatile

Once loaded, data is generally not continuously updated like an operational database.

Memory hook

💡 Key Insight

SITN

Subject-oriented

Integrated

Time-variant

Non-volatile

Database vs Data Warehouse

Database	Data Warehouse
Operational	Analytical
Current transactions	Historical data
OLTP	OLAP
Frequent updates	Mostly read/analysis
Normalized commonly	Often dimensional models

[UPLOADED PYQ — IT335 Mid]

Key Insight

How is a data warehouse different from a database? How are they similar?

4. Transactional Databases

A transactional database records individual business transactions.

Example:

```
T1 → {Bread, Milk}
T2 → {Milk, Eggs}
T3 → {Bread, Butter}
```

Important for later:

Key Insight

Association mining commonly operates on transaction databases.

5. Advanced Database Systems & Applications

Data mining can operate over different database types:

- relational databases
- object-oriented databases
- object-relational databases
- spatial databases
- temporal databases
- multimedia databases
- text databases
- Web databases
- heterogeneous databases
- distributed databases

Why this matters

Different data types require different mining techniques.

Example:

```
Images → image mining
Text → text mining
Web → Web mining
Spatial data → spatial mining
```

6. Data Mining Functionalities

Major functionalities:

```
Data Mining
├── Characterization
├── Discrimination
├── Association
├── Correlation
├── Classification
├── Prediction
├── Clustering
└── Outlier Analysis
```

Characterization

Produces a general summary of a target class.

Example:

💡 Key Insight

Describe students with $CGPA > 8$.

Discrimination

Compares one class against another.

Example:

💡 Key Insight

Compare students with $CGPA > 8$ against students with $CGPA \leq 8$.

[UPLOADED PYQ — IT426 Mid]

💡 Key Insight

Differentiate between Data Characterization and Data Discrimination with examples.

Association

Finds relationships among items.

Example:

Bread → Butter

Classification

Predicts a categorical class.

Example:

Email → Spam / Not Spam

Prediction

Predicts a value.

Example:

Past sales → Future sales

Clustering

Groups similar objects without predefined labels.



Outlier Analysis

Finds unusual objects.

Example:

💡 Key Insight

A credit-card transaction of ₹5,00,000 when the normal spending pattern is ₹2,000–₹10,000.

7. Classification of Data Mining Systems

Systems can be classified according to:

Data mined

- relational

- transactional
- spatial
- text
- multimedia
- Web

Knowledge discovered

- characterization
- association
- classification
- clustering
- outliers

Techniques used

- machine learning
- statistics
- database systems
- visualization
- pattern recognition

Applications

- banking
- healthcare
- retail
- telecom
- cybersecurity
- recommendation systems

8. Major Issues in Data Mining

Important issues:

Data quality

- missing values
- noise
- inconsistent values
- duplicate data

Scalability

Algorithm should work with huge datasets.

High dimensionality

Many attributes increase computational complexity.

Heterogeneous data

Different formats/sources.

Privacy and security

Sensitive information must be protected.

Pattern evaluation

Mining can produce huge numbers of patterns.

Need:

Key Insight

Interestingness measures

User interaction

Users may need:

- constraints
- visualization
- interactive mining

[UPLOADED PYQ — IT335 End]

Key Insight

Explain the functions and major issues in data mining.

9. Data

Data = collection of objects and their attributes.

Example:

Student	Age	CGPA	Department
A	20	8.5	CSE

Here:

- object = student
- attributes = age, CGPA, department

10. Data Objects & Attribute Types

Data Object

An entity represented by a collection of attributes.

Example:


```
Student = {Age, CGPA, Department}
```

Attribute Types

Nominal

Names/categories without ordering.

Example:

```
Department = {CSE, ECE, ME}
```

Binary

Two values.

```
Yes / No  
0 / 1
```

Ordinal

Ordered categories.

```
Poor < Average < Good < Excellent
```

Numeric

Numerical values.

Two common types:

Interval

Differences meaningful, zero not absolute.

Ratio

Both differences and ratios meaningful.

Example:

- temperature → interval
- weight → ratio

11. Basic Statistical Descriptions

Important measures:

Mean

$$\bar{x} = \frac{\sum x_i}{n}$$

Median

Middle value after sorting.

Mode

Most frequent value.

Range

$$Range = max - min$$

Variance

$$\sigma^2 = \frac{\sum (x_i - \bar{x})^2}{n}$$

Standard deviation

$$\sigma = \sqrt{\sigma^2}$$

Quartiles

Q1 → 25%
Q2 → 50% = Median
Q3 → 75%

Interquartile Range

$$IQR = Q3 - Q1$$

Five-number summary

Minimum
Q1
Median
Q3
Maximum

Box plot

Min — [Q1 | Median | Q3] — Max

12. Data Visualization

Visualization converts data into graphical form.

Common methods:

- bar chart
- histogram
- pie chart
- scatter plot
- box plot
- line graph

Scatter plot

Useful for detecting relationships between two numerical attributes.



13. Measuring Data Similarity

Similarity tells us:

💡 Key Insight

How alike are two objects?

Higher similarity = more alike.

Used in:

- clustering
- recommendation
- nearest-neighbor methods

Cosine similarity

For vectors:

$$\text{sim}(X, Y) = \frac{X \cdot Y}{\|X\| \|Y\|}$$

Common in text/vector data.

14. Measuring Data Dissimilarity

Dissimilarity tells us:

💡 Key Insight

How different are two objects?

For numerical data, common measure:

Euclidean distance

$$d(X, Y) = \sqrt{\sum_i (x_i - y_i)^2}$$

For two-dimensional points:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Manhattan distance

$$d(X, Y) = \sum_i |x_i - y_i|$$

Core relationship

Similarity ↑ → objects more alike
Dissimilarity ↑ → objects more different

M1 MEMORY PALACE

Think:

DATABASES
↓
FUNCTIONS
↓
DATA
↓
STATISTICS
↓
VISUALIZATION
↓
SIMILARITY

M1 PYQ set

Your uploaded papers repeatedly test:

- KDD process
- characterization vs discrimination
- mining functionality/issues
- preprocessing
- statistical measures
- database vs warehouse
- data warehouse concepts

MODULE II — DATA PROCESSING

6/6 topics

The uploaded syllabus lists **Cleaning, Integration, Transformation, Reduction, Discretization and Concept Hierarchy Generation.**

15. Data Cleaning

Real-world data is often:

Incomplete
Noisy
Inconsistent
Duplicate

Cleaning attempts to improve quality.

Missing Values

Methods:

Ignore tuple

Delete record if missing value is problematic.

Fill manually

Human provides value.

Global constant

Example:

Unknown

Mean

Replace numerical missing value by mean.

Median

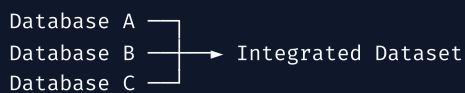
Useful when outliers exist.

Most probable value

Use statistical/model-based estimation.

16. Data Integration

Combines data from multiple sources.



Problems:

- schema integration
- naming conflicts
- redundancy
- duplicate attributes
- inconsistent values

Example:

DOB
Birth_Date
DateOfBirth

may represent the same attribute.

17. Data Transformation

Transforms data into suitable form.

Major techniques:

- smoothing
- aggregation

- generalization
- normalization
- attribute construction

Min-Max Normalization

Transforms x into $[new_min, new_max]$.

$$x' = \frac{x - min}{max - min}(new_max - new_min) + new_min$$

Example

Given:

200, 300, 400, 600, 1000

with target range:

10 to 20

For $x = 400$:

$$x' = \frac{400 - 200}{1000 - 200}(20 - 10) + 10 = 12.5$$

[UPLOADED PYQ] exactly asks this normalization problem.

Z-score normalization

$$x' = \frac{x - \mu}{\sigma}$$

When mean absolute deviation is specified, use the form required by the question/source convention.

18. Data Reduction

Goal:

💡 Key Insight

Reduce data size while preserving important information.

Techniques:

```
Data Reduction
├── Aggregation
├── Sampling
├── Dimensionality Reduction
├── Numerosity Reduction
└── Compression
```

Example

Instead of storing:

```
Daily sales
```

store:

```
Monthly sales
```

This is aggregation.

19. Data Discretization

Converts continuous values into intervals.

Example:

```
Age:
0-12    → Child
13-19   → Teen
20-59   → Adult
60+     → Senior
```

Useful for:

- classification
- rule generation
- reducing complexity

Methods include:

- binning
- histogram analysis
- clustering
- decision-tree-based discretization

20. Concept Hierarchy Generation

Creates higher-level concepts from lower-level values.

Example:

City
↓
State
↓
Country

Or:

Street
↓
City
↓
State
↓
Country

Useful for:

- data generalization
- OLAP
- concept-oriented mining

Methods

Concept hierarchies can be generated using:

- schema definitions
- partial ordering
- expert knowledge
- data-driven methods

[UPLOADED PYQ — IT426 End]

💡 Key Insight

Explain the significance of concept hierarchy for nominal data and two methods of generating it.

M2 MEMORY

Remember:

💡 Key Insight

C-I-T-R-D-H

Clean

Integrate

Transform

Reduce

Discretize

Hierarchy

MODULE III — DATA WAREHOUSE

9/9 topics

Your uploaded syllabus has **Basic Concepts, Modeling, Data Cube, OLAP, Design & Usage, Implementation, Attribute-Oriented Induction, Data Cube Computation and Preliminary Concepts.**

21. Basic Concepts of Data Warehouse

Think:

```
Operational Sources
  ↓
  ETL
  ↓
Data Warehouse
  ↓
  OLAP
  ↓
Analysis / Decision
```

ETL

Extract → Transform → Load

22. Data Warehouse Modeling

Three common models:

Star schema

Central fact table connected directly to dimension tables.

```
      Time
      |
Customer — FACT — Product
      |
      Location
```

Simple and fast for queries.

Snowflake schema

Dimensions are normalized.

```
      Time
      |
Customer — FACT — Product
  |           |
  Region     Category
```

More normalized, more joins.

Fact constellation

Multiple fact tables share dimensions.



Star vs Snowflake

Star	Snowflake
Less normalized	More normalized
Fewer joins	More joins
Simpler	More complex
Faster query access often	Can reduce redundancy
Easier to understand	More storage-efficient dimensions

[UPLOADED PYQ] asks this comparison directly.

23. Data Cube

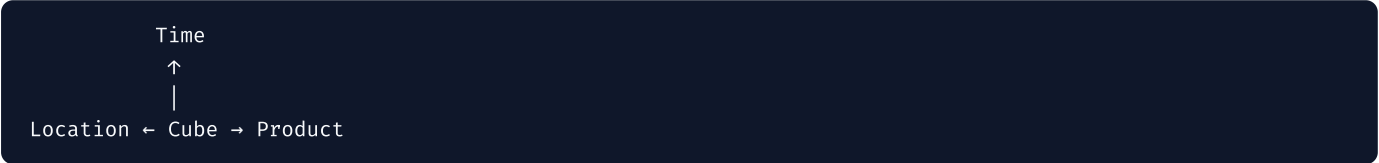
A data cube provides multidimensional analysis.

Example:

Dimensions:
Product
Time
Location

Measure:
Sales

Conceptually:



Cuboid

A cuboid represents data at a particular level of aggregation.

Base cuboid

Most detailed level.

Apex cuboid

Highest-level aggregation.

24. OLAP

Online Analytical Processing

Used for multidimensional analysis.

Important operations:

Roll-up

Move to higher-level aggregation.

City → State → Country

Drill-down

Move toward more detail.

Country → State → City

Slice

Fix one dimension.

Year = 2026

Dice

Select ranges/subsets across multiple dimensions.

Pivot

Rotate dimensions for another view.

OLTP vs OLAP

OLTP	OLAP
Transaction processing	Analysis
Current data	Historical/aggregated
Many short transactions	Complex queries
Insert/update/delete	Mostly read
Normalized commonly	Dimensional models

[UPLOADED PYQ] asks this repeatedly.

25. Data Warehouse Design & Usage

Important design perspectives:

Top-down

Start with enterprise warehouse.

Bottom-up

Build data marts and integrate them.

Hybrid

Combines both approaches.

Design must consider:

- business requirements
- dimensions
- measures
- granularity
- source systems
- ETL
- performance

[UPLOADED PYQ — IT426 Mid]

Key Insight

What different views must be considered while designing a data warehouse?

26. Data Warehouse Implementation

Major stages:

Source Systems



Extraction



Cleaning



Transformation



Loading



Warehouse



OLAP / Mining

Important implementation concerns:

- indexing
- partitioning
- materialized views
- join indices
- refresh
- metadata

27. Attribute–Oriented Induction

AOI summarizes data by:

1. Removing irrelevant attributes.
2. Generalizing remaining attributes using concept hierarchies.
3. Aggregating identical generalized tuples.
4. Presenting generalized knowledge.

Example:

Ranchi
Patna
Jamshedpur

may generalize into:

Jharkhand
Bihar

depending on hierarchy.

28. Data Cube Computation

If there are dimensions:

A, B, C

different cuboids can be created:

```
ABC
AB
AC
BC
A
B
C
∅
```

For n dimensions, the full cube has:

$$2^n$$

cuboids.

Materialization choices

Full materialization

Compute all cuboids.

No materialization

Compute only when requested.

Partial materialization

Compute selected cuboids.

[UPLOADED PYQ] asks the choices of data cube materialization given a base cuboid.

29. Preliminary Concepts

Important warehouse terminology:

Fact table

Contains:

- measures
- foreign keys to dimensions

Dimension table

Describes dimensions.

Measure

Numeric quantity being analyzed.

Example:

```
Sales = ₹50,000
Quantity = 500
```

Granularity

Level of detail.

Example:

Daily → Monthly → Yearly

Lower granularity generally means more summarized information.

M3 PYQ CLUSTER

Repeated themes:

- data warehouse vs database
- OLTP vs OLAP
- warehouse architecture/design
- star vs snowflake
- data cube
- cube materialization
- concept hierarchy
- normalization
- warehouse implementation

MODULE IV — FREQUENT PATTERN MINING

7/7 topics

Your syllabus specifies **basic frequent pattern mining**, **associations**, **correlations**, **frequent-itemset methods**, **Apriori**, **FP-Growth** and **interesting-pattern evaluation**.

This is one of the **highest-value modules** because it combines theory + numerical algorithms.

30. Basic Concepts of Frequent Pattern Mining

A **frequent pattern** is a pattern occurring frequently in a dataset.

Example transactions:

T1 = Bread, Milk
T2 = Bread, Butter
T3 = Bread, Milk, Butter

Bread is frequent.

Bread + Milk may also be frequent.

31. Associations

Association rule:

$$X \rightarrow Y$$

means:

💡 Key Insight

When X occurs, Y tends to occur.

Example:

Bread → Butter

Support

$$Support(X) = \frac{\text{transactions containing } X}{\text{total transactions}}$$

For rule:

$$Support(X \rightarrow Y) = Support(X \cup Y)$$

Confidence

$$Confidence(X \rightarrow Y) = \frac{Support(X \cup Y)}{Support(X)}$$

Interpretation:

💡 Key Insight

Among transactions containing X, what fraction also contains Y?

Example

Suppose:

100 transactions
40 contain Bread
30 contain Bread + Butter

Then:

$$\text{Support}(Bread \rightarrow Butter) = 30\%$$

$$\text{Confidence} = 30/40 = 75\%$$

32. Correlations

Support and confidence alone may sometimes be misleading.

Correlation measures whether occurrence of X and Y is actually associated.

Lift

$$\text{Lift}(X \rightarrow Y) = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

Interpretation:

Lift > 1 → positive association
Lift = 1 → independent
Lift < 1 → negative association

[UPLOADED PYQ — IT426 Mid]

💡 Key Insight

Compare the significance of Support and Confidence in association analysis.

33. Frequent Itemset Mining Methods

General process:

```
Database
↓
Generate candidates
↓
Count support
↓
Keep frequent patterns
↓
Generate rules
```

Main approaches:

Apriori

Candidate generation + pruning.

FP-Growth

Pattern-growth approach.

34. Apriori Algorithm

The central principle:

💡 Key Insight

If an itemset is frequent, all of its non-empty subsets must also be frequent.

Equivalent pruning rule:

💡 Key Insight

If an itemset is infrequent, every larger itemset containing it is also infrequent.

This is the **Apriori property**.

Algorithm

```
Find frequent 1-itemsets
    ↓
Generate candidate 2-itemsets
    ↓
Prune
    ↓
Count support
    ↓
Find frequent 2-itemsets
    ↓
Generate candidates
    ↓
Repeat
```

Join Step

Combine frequent $(k-1)$ -itemsets to generate k -itemset candidates.

Prune Step

Remove candidate if any $(k-1)$ subset is not frequent.

[UPLOADED PYQ — IT426 End]

💡 Key Insight

Explain hash-based and partitioning techniques for improving Apriori and illustrate join/prune steps.

35. FP-Growth

FP-Growth avoids explicit candidate generation.

Main idea:

```
Database
↓
Frequency ordering
↓
FP-tree
↓
Conditional pattern bases
↓
Conditional FP-trees
↓
Frequent patterns
```

Advantages over Apriori

- avoids huge candidate sets
- often requires fewer database scans
- compact representation

Uploaded PYQ — FP-Growth

Your IT335 End paper gives:

```
min support = 60%
min confidence = 50%
```

Transactions:

```
10: Beer, Nuts, Diapers
20: Beer, Coffee, Diapers, Nuts
30: Beer, Diapers, Eggs
40: Beer, Nuts, Eggs, Milk
50: Nuts, Coffee, Diapers, Eggs, Milk
```

and asks for frequent patterns using FP-Growth.

This is a **must-solve numerical**.

36. Interesting Pattern Evaluation

Not every discovered pattern is useful.

Measures include:

Support

Frequency.

Confidence

Reliability of rule.

Lift

Strength relative to independence.

Other concepts may include:

- novelty
- usefulness
- unexpectedness
- statistical significance

Mental model

Frequent $\neq$ Automatically Interesting



ASSOCIATION RULE MEMORY

Remember:

💡 Key Insight

S-C-L

Support → How often?

Confidence → How reliable?

Lift → How much stronger than chance?

MODULE V — ADVANCED PATTERN MINING

10/10 topics

The uploaded syllabus lists **Pattern Mining Road Map**, **Multilevel**, **Multidimensional**, **Constraint-Based**, **High-Dimensional**, **Colossal**, **Compressed**, **Approximate patterns**, **Pattern Exploration** and **Pattern Applications**.

37. Pattern Mining Road Map

Think of pattern mining as progressing from simple to increasingly complex patterns:

```
Frequent Itemsets
  ↓
Multilevel
  ↓
Multidimensional
  ↓
Constraint-Based
  ↓
High-Dimensional
  ↓
Compressed / Approximate
  ↓
Pattern Exploration
  ↓
Applications
```

Goal:

💡 Key Insight

Find useful patterns without generating an overwhelming number of irrelevant patterns.

38. Multilevel Pattern Mining

Uses concept hierarchies.

Example:

```
Milk
  ↓
Dairy
  ↓
Food
```

At lower level:

```
Milk → Bread
```

At higher level:

```
Dairy → Bakery
```

Why?

Patterns at different abstraction levels can reveal different knowledge.

39. Multidimensional Pattern Mining

Patterns involve multiple attributes rather than only items.

Example:

```
Age = 20-30
AND
Income = High
AND
Buys = Laptop
```

Dimensions:

- age
- income
- product

Contrast

Single-dimensional

```
buys(X, Laptop) → buys(X, Mouse)
```

Multidimensional

```
age + income + product
```

40. Constraint-Based Frequent Pattern Mining

Instead of mining everything, specify constraints.

Example:

💡 Key Insight

Find products where total price > ₹5,000.

Possible constraints:

- minimum support
- item constraints
- aggregate constraints
- length constraints
- value constraints

Benefit

Search space becomes smaller.

```
All Patterns
  ↓
Constraints
  ↓
Relevant Patterns
```

41. Mining High-Dimensional Data

High-dimensional dataset:

$n \text{ objects} \times d \text{ attributes}$

where d is large.

Problems:

- computational complexity
- sparse data
- irrelevant attributes
- curse of dimensionality

Approaches include:

- feature selection
- dimensionality reduction
- subspace methods
- specialized pattern representations

42. Mining Colossal Patterns

A **colossal pattern** is a frequent pattern that has no frequent proper superset.

It represents an extremely large maximal frequent structure.

Why useful?

Instead of storing enormous numbers of smaller patterns, colossal patterns can summarize important frequent structures.

43. Mining Compressed Patterns

Goal:

Key Insight

Represent many frequent patterns compactly.

Important concepts include:

Closed frequent pattern

A frequent pattern is closed if none of its immediate/superset patterns has the same support.

Maximal frequent pattern

A frequent pattern with no frequent superset.

Relationship

```
All frequent patterns
      ↓
Closed patterns
      ↓
Maximal patterns
```

Maximal patterns provide stronger compression but lose support information for some subsets.

44. Mining Approximate Patterns

Exact patterns may be too restrictive in noisy/large datasets.

Approximate mining allows:

- similarity
- tolerance
- probabilistic matching
- approximate frequency

Useful where data is:

- noisy
- uncertain
- continuous
- huge

45. Pattern Exploration

Finding patterns is only half the problem.

We must explore them.

Methods:

- visualization
- filtering
- ranking
- drill-down
- user-defined constraints
- interestingness measures

Pipeline:

```
Mining
↓
Thousands/Millions Patterns
↓
Filter
↓
Rank
↓
Visualize
↓
Human Insight
```

46. Pattern Applications

Frequent/advanced patterns are used in:

Retail

Market basket analysis.

Recommendation

Users/items with similar behavior.

Web mining

User navigation patterns.

Healthcare

Disease/diagnostic patterns.

Finance

Fraud and transaction patterns.

Security

Attack patterns.

Bioinformatics

Gene/protein relationships.

THE OTHER MAJOR DMCT BLOCK: CLASSIFICATION & CLUSTERING

There is an important point here.

Your **46–topic CS24303 outline** groups the official syllabus as above, but your uploaded **IT335/IT426 examination papers also repeatedly test classification and clustering**, especially:

- Backpropagation
- accuracy

- precision
- recall
- error rate
- K-Means
- DBSCAN
- Naïve Bayes
- Sequential Covering
- PAM
- BIRCH.

So I would **NOT omit these from your exam preparation**, even though they are not separately visible in the 46-item outline returned from your study-guide file.

CLASSIFICATION MASTER BLOCK

Classification

Supervised learning.

```

Training Data
  ↓
Learn Model
  ↓
New Object
  ↓
Predicted Class
  
```

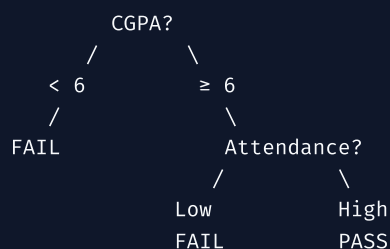
Example:

```

Student attributes
  ↓
Pass / Fail
  
```

Decision Tree

Tree structure:



Important concepts

- root

- internal node
- branch
- leaf
- splitting attribute

Common splitting measures:

Information Gain

$$IG(S, A) = Entropy(S) - \sum_v \frac{|S_v|}{|S|} Entropy(S_v)$$

Entropy

$$Entropy(S) = - \sum_i p_i \log_2 p_i$$

Gini Index

$$Gini(S) = 1 - \sum_i p_i^2$$

Naïve Bayes

Based on Bayes theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Naïve assumption:

Key Insight

Attributes are conditionally independent given the class.

Thus:

$$P(X|C) = \prod_i P(x_i|C)$$

Procedure

```
Calculate class probabilities
  ↓
Calculate conditional probabilities
  ↓
Apply Bayes
  ↓
Compare classes
  ↓
Choose highest probability
```

[UPLOADED PYQ — IT426 End]

💡 Key Insight

Write steps of Naïve Bayesian classification and examine the naïve class-conditional independence assumption.

Rule-Based Classification

Rules:

```
IF condition
THEN class
```

Example:

```
IF CGPA > 8
AND attendance > 75%
THEN PASS
```

Sequential Covering

General idea:

```
Start with all training examples
  ↓
Find good rule
  ↓
Remove covered examples
  ↓
Find next rule
  ↓
Repeat
```

[UPLOADED PYQ] asks for a Sequential Covering rule-induction algorithm.

Backpropagation

Used to train neural networks.

Core idea:

```
Input
↓
Forward propagation
↓
Prediction
↓
Error
↓
Backward propagation
↓
Update weights
↓
Repeat
```

For a neuron:

$$net = \sum_i w_i x_i + b$$

$$y = f(net)$$

Error:

$$E = \frac{1}{2}(t - y)^2$$

Weights are updated using gradient-based learning.

[UPLOADED PYQ — IT335 End]

💡 Key Insight

Explain Backpropagation and calculate the Backpropagation error for given values.

CLASSIFICATION METRICS

Confusion matrix:

	Predicted +	Predicted -
Actual +	TP	FN
Actual -	FP	TN

Accuracy

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$Precision = \frac{TP}{TP + FP}$$

Recall

$$Recall = \frac{TP}{TP + FN}$$

Error rate

$$Error = \frac{FP + FN}{Total}$$

[UPLOADED PYQ] specifically asks accuracy, precision, error rate and recall.

Memory

💡 Key Insight

Precision = predicted positives → how many correct?

💡 Key Insight

Recall = actual positives → how many found?



CLUSTERING MASTER BLOCK

Clustering = unsupervised grouping.

```
Raw Objects
  ↓
Similarity / Distance
  ↓
Clusters
```

K-MEANS

Partitioning method.

Given k clusters:

Step 1

Choose k centroids.

Step 2

Assign each point to nearest centroid.

Step 3

Recalculate centroid.

$$\mu_j = \frac{1}{|C_j|} \sum_{x_i \in C_j} x_i$$

Step 4

Repeat until assignments stabilize.

```
Choose centroids
  ↓
Assign
  ↓
Recalculate
  ↓
Repeat
```

Objective

Minimize within-cluster squared error:

$$SSE = \sum_{j=1}^k \sum_{x \in C_j} ||x - \mu_j||^2$$

[UPLOADED PYQ]

💡 Key Insight

What is partitioning method? Explain K-Means with suitable example.

PAM — Partitioning Around Medoids

Unlike K-Means, PAM uses an actual data object as cluster representative.

That object is the **medoid**.

Basic steps

```
Choose k medoids
↓
Assign objects
↓
Try replacing medoid
↓
Calculate cost
↓
Keep beneficial swap
↓
Repeat
```

K-Means vs PAM

K-Means	PAM
Centroid may not be data object	Medoid is actual object
Sensitive to outliers	More robust
Usually faster	More computationally expensive
Mean-based	Representative-object based

[UPLOADED PYQ] asks PAM and why it can be superior to K-Means.

DBSCAN

Density-based clustering.

Two main parameters:

- ϵ — neighborhood radius
- MinPts — minimum number of points

Point types

Core

Enough neighboring points.

Border

Not core itself but lies near core point.

Noise

Doesn't belong to a sufficiently dense region.

```
Dense region      Noise
● ● ● ●          ×
● ● ● ●
● ● ●
```

Advantages

- discovers arbitrary-shaped clusters

- identifies noise
- doesn't require k in advance

Limitation

Can struggle when clusters have very different densities.

[UPLOADED PYQ] directly asks DBSCAN.

BIRCH

Balanced Iterative Reducing and Clustering using Hierarchies

Designed for large datasets.

Central structure:

💡 Key Insight

CF-tree — Clustering Feature Tree

Clustering Feature:

$$CF = (N, LS, SS)$$

where:

- N = number of points
- LS = linear sum
- SS = squared sum

Phases

Conceptually:

```
Build CF-tree
  ↓
Condense / refine
  ↓
Cluster subclusters
  ↓
Optional refinement
```

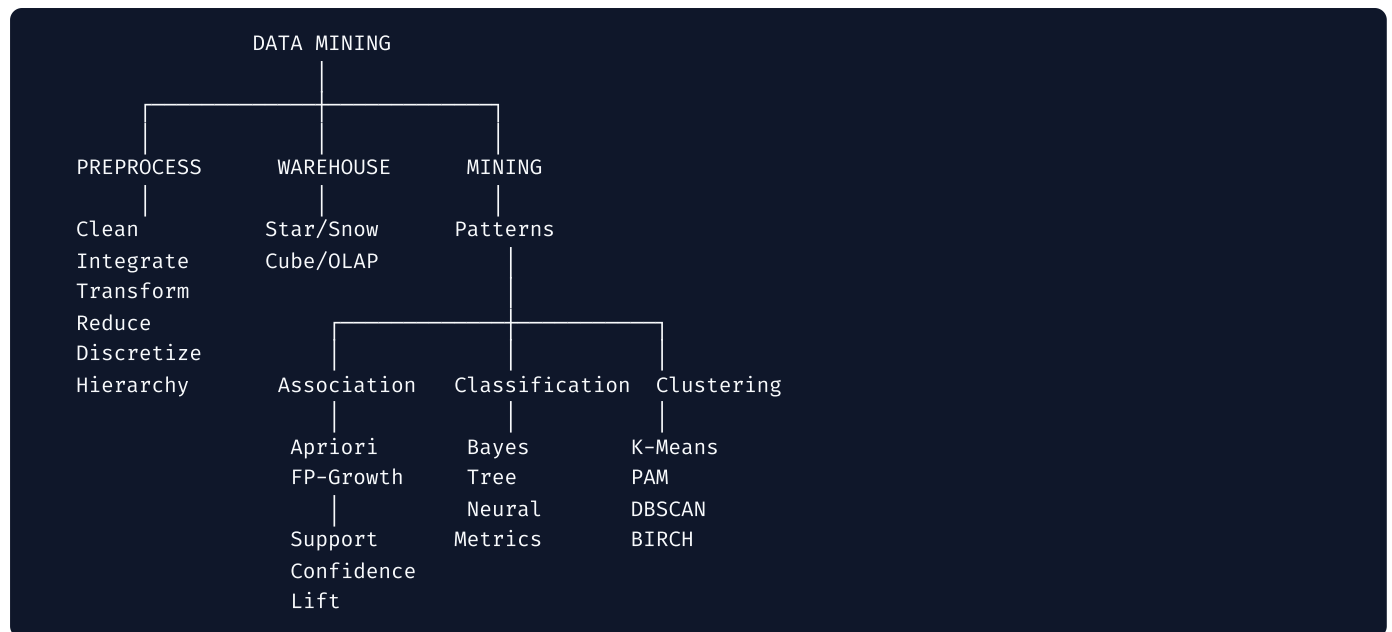
Why useful?

Instead of storing every point for every clustering operation, BIRCH maintains compact summaries.

[UPLOADED PYQ] asks for the primary BIRCH phases and how CF-tree addresses scalability.

🔥 DMCT ALGORITHM MAP

This is the part I want you to memorize conceptually rather than as isolated definitions:



COMPLETE 46–TOPIC CHECKLIST

M1 — Introduction to Data Mining — 14

- ☐ Introduction to Data Mining
- ☐ Relational Databases
- ☐ Data Warehouses
- ☐ Transactional Databases
- ☐ Advanced Database Systems and Applications
- ☐ Data Mining Functionalities
- ☐ Classification of Data Mining Systems
- ☐ Major Issues in Data Mining
- ☐ Data
- ☐ Data Objects and Attribute Types
- ☐ Basic Statistical Descriptions of Data
- ☐ Data Visualization
- ☐ Measuring Data Similarity
- ☐ Measuring Data Dissimilarity

M2 — Data Processing — 6

- ☐ Data Cleaning
- ☐ Data Integration
- ☐ Data Transformation
- ☐ Data Reduction
- ☐ Data Discretization
- ☐ Concept Hierarchy Generation

M3 — Data Warehouse — 9

- ☐ Basic Concepts of Data Warehouse
- ☐ Data Warehouse Modeling
- ☐ Data Cube
- ☐ OLAP
- ☐ Data Warehouse Design and Usage
- ☐ Data Warehouse Implementation
- ☐ Data Generalization by Attribute-Oriented Induction
- ☐ Data Cube Computation
- ☐ Preliminary Concepts

M4 — Frequent Pattern Mining — 7

- ☐ Basic Concepts
- ☐ Associations
- ☐ Correlations
- ☐ Frequent Itemset Mining Methods
- ☐ Apriori
- ☐ FP-Growth
- ☐ Interesting Pattern Evaluation

M5 — Advanced Pattern Mining — 10

- ☐ Pattern Mining Road Map
- ☐ Multilevel Pattern Mining
- ☐ Multidimensional Pattern Mining
- ☐ Constraint-Based Frequent Pattern Mining
- ☐ Mining High-Dimensional Data
- ☐ Mining Colossal Patterns
- ☐ Mining Compressed Patterns
- ☐ Mining Approximate Patterns
- ☐ Pattern Exploration
- ☐ Pattern Applications

46/46 syllabus topics.



BUT FOR YOUR EXAM: ADD THESE PYQ-CRITICAL TOPICS

Because your actual uploaded IT335/IT426 papers repeatedly ask them:

```
Classification
├── Decision Tree
├── Naïve Bayes
├── Rule-Based Classification
├── Sequential Covering
├── Backpropagation
└── Accuracy / Precision / Recall / Error
```

```
Clustering
├── K-Means
├── PAM
├── DBSCAN
└── BIRCH
```

IT335 End explicitly asks **Backpropagation, classification metrics, K-Means and DBSCAN**.

IT426 End additionally asks **regression vs classification, Sequential Covering, Naïve Bayes, PAM and BIRCH**.

So I would treat these as **exam-critical DMCT material**, not skip them merely because the 46-item checklist doesn't list each under a separate heading.

NEUROSCIENCE-BASED DMCT STUDY METHOD

For DMCT, don't use passive rereading.

Use this cycle:

1. Predict

Before learning Apriori, ask:

Key Insight

"If a 3-itemset contains an infrequent 2-itemset, can it ever be frequent?"

Answer:

No.

That's retrieval before explanation.

2. Explain

After studying K-Means, close the notes.

Explain:

Key Insight

"Why does K-Means recalculate the centroid?"

If you can't explain it simply, you don't own it yet.

3. Generate

Don't look at the solution.

Take:

```
Transactions
↓
Calculate support
↓
Find frequent itemsets
↓
Generate rules
```

4. Interleave

Don't solve 10 Apriori questions consecutively.

Instead:

```
Apriori
↓
Normalization
↓
K-Means
↓
OLAP
↓
Naïve Bayes
↓
FP-Growth
↓
Statistics
↓
DBSCAN
```

This forces your brain to identify **which algorithm applies**, rather than simply repeating a procedure.

ALGORITHM MEMORY TECHNIQUE

Apriori

Key Insight

Generate → Join → Prune → Count

FP-Growth

Key Insight

Compress → Grow → Mine

K-Means

💡 Key Insight

Assign → Average → Repeat

PAM

💡 Key Insight

Medoid → Assign → Swap → Cost → Repeat

DBSCAN

💡 Key Insight

Core → Expand → Border → Noise

BIRCH

💡 Key Insight

Compress → Cluster → Refine

Naïve Bayes

💡 Key Insight

Prior → Conditional → Multiply → Compare

Backpropagation

💡 Key Insight

Forward → Error → Backward → Update

These short chains are your **retrieval cues**.

DMCT HIGH-YIELD NUMERICALS

You should be able to solve these without notes:



1. Min-Max normalization
2. Z-score normalization
3. Mean/median/variance/standard deviation
4. Euclidean/Manhattan distance
5. Entropy
6. Information Gain
7. Gini Index
8. Support

- 9. Confidence
- 10. Lift
- 11. Apriori candidate generation
- 12. FP-Growth
- 13. Naïve Bayes
- 14. K-Means iterations
- 15. Backpropagation
- 16. Confusion-matrix metrics
- 17. PAM
- 18. DBSCAN ϵ /MinPts reasoning
- 19. Data-cube materialization

Your uploaded papers specifically demonstrate normalization, FP-Growth and backpropagation as calculation-oriented questions.

MOST IMPORTANT PYQs FROM YOUR FILES

M1/M2/M3

- Data warehouse vs database
- KDD process
- Mining challenges/functions
- Statistical measures of dispersion
- Preprocessing techniques
- Data cube
- Snowflake schema
- OLTP vs OLAP
- Warehouse architecture
- Normalization
- Concept hierarchy
- Star vs Snowflake
- Cube materialization

M4

- Apriori algorithm
- Apriori join/prune
- Hash-based Apriori
- Partition-based Apriori
- FP-Growth
- Support vs confidence
- FP-Growth numerical with `minsup=60%` , `minconf=50%`

Classification

- Backpropagation
- Accuracy

- Precision
- Recall
- Error rate
- Naïve Bayes
- Sequential Covering
- Classification vs regression

Clustering

- K-Means
- DBSCAN
- PAM
- BIRCH
- CF-tree
- Partitioning methods

FINAL DMCT MASTER MAP

If you remember only this structure:

M1

DATA

- Databases
- Warehouses
- Mining Functions
- Data Types
- Statistics
- Visualization
- Similarity

M2

PREPROCESS

- Clean
- Integrate
- Transform
- Reduce
- Discretize
- Hierarchy

M3

WAREHOUSE

- Modeling
- Star/Snowflake
- Cube
- OLAP
- Design
- Implementation
- AOI
- Materialization

M4

PATTERNS

- Association
- Support
- Confidence
- Correlation/Lift
- Apriori
- FP-Growth

M5

ADVANCED PATTERNS

- Multilevel
- Multidimensional
- Constraints
- High-dimensional
- Colossal
- Compressed
- Approximate
- Exploration/Application

PYQ CRITICAL

- Classification
 - Decision Tree
 - Naïve Bayes
 - Backpropagation
 - Metrics
- Clustering

- K-Means
- PAM
- DBSCAN
- BIRCH

This is the version I would study from for your DMCT exam: the **46 official syllabus topics + the classification/clustering material that demonstrably appears in your uploaded PYQs**, rather than blindly relying only on the checklist.

And for DMCT, the highest-return strategy is:

 Key Insight

Understand → derive → solve → explain from memory → solve PYQ → spaced recall.

That will be substantially more effective than simply reading these notes repeatedly.